

Exercise 2

Every two-element set of connectives containing $\neg$ and one of $\{\wedge, \vee, \rightarrow\}$ is a functionally complete subset of (or in other words can be used to define) $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$.

Part 1: Define $\wedge$, $\rightarrow$, and $\leftrightarrow$ using only $\vee$ and $\neg$.

Answer

$$\begin{aligned} p \wedge q &:= \neg(\neg p \vee \neg q), \\ p \rightarrow q &:= \neg p \vee q, \\ p \leftrightarrow q &:= \neg(\neg p \vee \neg q) \vee \neg(p \vee q) \end{aligned}$$

Part 2: Define exclusive disjunction (XOR, $\underline{\vee}$) using only $\wedge$ and $\neg$.

Answer

$$p \underline{\vee} q := \neg(\neg p \wedge \neg q) \wedge \neg(p \wedge q)$$

Part 3: Actually, there exists two binary operators (dual to each other), NAND and NOR, that are complete by itself. The NAND operators means "not both" and can be defined as follows: $p \uparrow q \leftrightarrow \neg(p \wedge q)$. The NOR operators means "neither nor" and can be defined as follows: $p \downarrow q \leftrightarrow \neg(p \vee q)$. Pick one of the operators and show that it is enough to define all the usual operators $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$.

Answer

$$\begin{aligned} \neg p &:= p \uparrow p, \\ p \wedge q &:= (p \uparrow q) \uparrow (p \uparrow q), \\ p \vee q &:= (p \uparrow p) \uparrow (q \uparrow q), \\ p \rightarrow q &:= p \uparrow (q \uparrow q), \\ p \leftrightarrow q &:= (p \uparrow (q \uparrow q)) \uparrow (p \uparrow (q \uparrow q)) \uparrow (q \uparrow (p \uparrow p)) \uparrow (q \uparrow (p \uparrow p)) \end{aligned}$$